CATEGORY THEORY ABSTRACTIONS FOR tnreason

RESEARCH NOTES IN THE ENEXA AND QROM PROJECTS

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ABSTRACT

We investigate the abstractions of tensor networks by string diagrams of categories for usage in the tnreason formalism.

1 Semiring contractions as categories

Each semiring $\mathbf{S} = (S, \oplus, \otimes, \mathbf{0}, \mathbf{1})$ defines a category:

- Objects are the elements of the cyclic group
- Morphisms are maps from the objects to $\mathbf{S}$

Open question: Are monads the property needed for message passing?

The contraction category on a semiring $\mathbf{S}$ is a symmetric monoidal category, where the functor is the tensor product.

1.0.1 Properties

Lemma 1. *Each contraction category on a semiring* $\mathbf{S}$ *is a cd-category. The copy morphism is the delta tensor on that semiring, and the discarding morphism the vector of coordinates* $\mathbf{1}$.

Lemma 2. *Each contraction category on a semiring* $\mathbf{S}$ *has a cap by the delta tensor seen as an effect.*

2 Normalization

Directed tensors are normalizations in the Mat category. We highlight them in the tnreason formalism by directed hyperedges.

When choosing the CPM functor mapping Hilb to Quant, quantum states are the normalized pure density matrices and unitaries channels.

3 Encoding schemes as functors

- Basis encodings, selection encodings as functors from the FinSet category to the Mat category
- Computation circuits and activation circuits as functors from the FinSet category to the Hilb category

3.1 Basis encodings as the $\mathrm{L}_{\mathbb{B}}$ functor

The category FinSet is defined as:

- Objects are finite sets
- Morphisms are total functions between them

The category $\mathrm{Mat}_{\mathbb{B}}$ of Boolean tensors is defined as:

- Objects are cyclic groups
- Morphisms are Boolean tensors

The basis encoding scheme is the $\mathrm{L}_{\mathbb{B}}$ functor:

- Finite sets $\mathcal{U}$ are mapped to cyclic groups $\mathbb{Z}_{|\mathcal{U}|}$
- Functions are mapped to Boolean tensors

Basis encodings are always deterministic morphisms: The contraction with the copy morphism (the delta tensor) is equal to the contraction of two basis encodings with different head variables.

There are different functors mapping Boolean matrices to further categories. Functors into any contraction category on a semiring just replace the 0 and 1 coordinates of the morphism by the $\mathbf{0}$ and $\mathbf{1}$ elements of the semiring.

3.2 Quantum measurement as functors

Computational basis measurement of circuits is a functor from quantum circuits to classical probability distributions. We can understand the functor based on contractions representing the Born rule. The measurement can be understood as a composition of the CPM functor with the dephasing function mapping further to $\mathrm{FinStoch}$.

References